

Degeneracy: $n=1$, 3
 $n=2$ 6

$$n=3 = n_x + n_y + n_z$$

$$\left[\begin{array}{ccc} 0 & 0 & 3 \\ 0 & 3 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 0 \\ 2 & 0 & 2 \\ 2 & 0 & 1 \\ 2 & 1 & 0 \\ 3 & 0 & 0 \end{array} \right]$$

10

missed one

Thus, the degeneracy is not $3n$. So, how do we find it?

$g(n)$: $g(1) = 3$

$$g(2) = 6$$

$$g(3) = 10$$

Two Methods: (i) Continues with our above

and assume $g(n)$ is a polynomial

Method I:

$$g(n) = \alpha + \beta n + \gamma n^2$$

Why only up to n^2 ? Well note that

$$g(1) = 3 = \alpha + \beta + \gamma$$

$$g(2) = 6 = \alpha + 2\beta + 4\gamma$$

and so, since $\alpha + \beta + \gamma = 3$ then we cannot have terms higher than n^2

we also have

$$g(3) = 10 = \alpha + 3\beta + 9\gamma$$

• we have 3-equations and 3 unknowns. we find:

$$\alpha = 1$$

$$\beta = \frac{3}{2}$$

$$\gamma = \frac{1}{2}$$

$$\begin{aligned} g(n) &= 1 + \frac{3}{2}n + \frac{1}{2}n^2 \\ &= \frac{1}{2}(2 + 3n + n^2) \\ &= \frac{1}{2}(n+1)(n+2) \end{aligned}$$

* Method II: $n = n_x + n_y + n_z$, for fixed n

• for fixed n_z : $n_x + n_y = \frac{n - n_z}{\text{fixed}}$

So we just have to figure out how many combinations of n_x and n_y we can make

$$\begin{aligned} n_x = 0: & n_y = n - n_z \\ n_x = 1: & n_y = n - n_z - 1 \\ n_x = 2: & n_y = n - n_z - 2 \\ & \vdots \end{aligned} \left. \vphantom{\begin{aligned} n_x = 0: \\ n_x = 1: \\ n_x = 2: \end{aligned}} \right\} \frac{n - n_z}{2} \text{ (terms)}$$

$$n_x = n_y: n_y = \frac{n - n_z}{2} \longrightarrow 1 \text{ - (term)}$$

$$\begin{aligned} n_x = n_y + 1: & n_y = \frac{n - n_z - 1}{2} \\ & \vdots \end{aligned} \left. \vphantom{\begin{aligned} n_x = n_y + 1: \\ \vdots \end{aligned}} \right\} \frac{n - n_z}{2} \text{ (terms)}$$

$$n_x = n - n_z: n_y = 0$$

$$(\text{total}) = \frac{n - n_z}{2} + 1 + \frac{n - n_z}{2} = \underline{n - n_z + 1}$$

* Note this was for fixed n_2 , so we must sum terms over n_2 up to n :

$$g_n = \sum_{n_2=0}^n (n - n_2 + 1) = \underbrace{\sum_{n_2=0}^n (n+1)}_{(n+1)(n+1)} - \underbrace{\sum_{n_2=0}^n n_2}_{\frac{n(n+1)}{2}}$$

$$g_n = (n+1)(n+1) - \frac{n(n+1)}{2}$$

$$= (n+1) \left(n+1 - \frac{n}{2} \right) = (n+1) \left(\frac{2n+2-n}{2} \right) = \frac{(n+1)(n+2)}{2}$$

$$g_n = \frac{(n+1)(n+2)}{2}$$

just as above.

→ Again, sorry about the error!



